

Research.md

Research & Innovation Strategy

Project: Bloom

Document: Research Vision and Technical Research Roadmap

Version: 1.0

Status: Living Research Document

1. Purpose

This document defines the long-term research direction of Bloom.

Unlike the Product Roadmap, which focuses on building features, this document focuses on answering research questions that can significantly improve Bloom's Learning Intelligence.

The objective is to continuously discover better ways to help people learn.

2. Research Philosophy

Bloom is built on one fundamental belief:

Learning is an optimization problem, not a content problem.

The internet already contains enormous amounts of educational content.

The research challenge is understanding:

- what to learn
 - when to learn
 - how to learn
 - from whom to learn
 - why learners succeed or fail
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3. Core Research Areas

Bloom's research is organized into seven domains.

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graph TD; A[Learning Science] --> B[Knowledge Modeling]; B --> C[Recommendation Systems]; C --> D[Creator Intelligence]; D --> E[Content Intelligence]; E --> F[Human-AI Learning]; F --> G[Learning Analytics];
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Learning Science
|
Knowledge Modeling
|
Recommendation Systems
|
Creator Intelligence
|
Content Intelligence
|
Human-AI Learning
|
Learning Analytics

4. Research Domain 1 — Learning Science

Goal

Understand how people actually learn.

Research Questions

- Why do learners quit?
- What makes learning enjoyable?
- How long should learning sessions be?
- How frequently should concepts be revised?
- How much difficulty is optimal?
- How does motivation change over time?

Potential Topics

- Cognitive Load
- Active Recall
- Spaced Repetition
- Interleaving
- Deliberate Practice
- Project-Based Learning
- Metacognition

- Flow State
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5. Research Domain 2 — Knowledge Representation

Goal

Model human knowledge computationally.

Research Questions

- How should concepts be represented?
- What defines a prerequisite?
- Can multiple valid learning paths exist?
- How should career knowledge be modeled?
- How do skills evolve?

Research Outputs

- Skill Graph
 - Career Graph
 - Concept Graph
 - Knowledge Dependencies
 - Learning Path Graphs
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6. Research Domain 3 — Recommendation Systems

Goal

Move beyond engagement-based recommendation systems.

Traditional recommendation systems optimize:

- Clicks
- Watch Time
- Revenue

Bloom researches recommendation systems that optimize:

- Learning efficiency
- Knowledge retention
- Skill mastery
- Career progression

Research Questions

- What is the next best learning experience?
- How should recommendation confidence be measured?
- How should uncertainty be handled?
- When should exploration occur instead of exploitation?

Potential Algorithms

- Knowledge Graph Traversal
- Reinforcement Learning
- Contextual Bandits
- Hybrid Recommendation Systems
- Multi-objective Optimization

7. Research Domain 4 — Creator Intelligence

Goal

Understand educators.

Research Questions

- What makes a creator effective?
- Which teaching styles suit different learners?
- How should explanation quality be measured?
- How does storytelling influence understanding?

Possible Creator Dimensions

- Teaching Style
 - Practical Focus
 - Beginner Friendliness
 - Communication Clarity
 - Technical Depth
 - Visual Quality
 - Engagement Quality
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8. Research Domain 5 — Content Intelligence

Goal

Evaluate educational resources objectively.

Research Questions

- What defines educational quality?
- How should outdated content be detected?
- How should duplicate resources be handled?
- How can concepts be automatically extracted?

Possible Signals

- Freshness
 - Accuracy
 - Completeness
 - Structure
 - Community Trust
 - Creator Reputation
 - Learning Outcomes
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9. Research Domain 6 — Human-AI Learning

Goal

Study collaboration between learners and AI.

Research Questions

- When should AI guide?
- When should AI stay silent?
- How much personalization is beneficial?
- Can AI detect learner frustration?
- Can AI encourage curiosity?

Research Areas

- Explainable AI
- AI Mentorship
- Conversational Learning
- Adaptive Learning
- Human Decision Support

10. Research Domain 7 — Learning Analytics

Goal

Understand learning at scale.

Research Questions

- Which concepts cause the highest dropout?
- Which creators produce the strongest learning outcomes?
- What predicts long-term success?
- How do learning habits evolve?

Potential Analytics

- Skill Progression
- Learning Velocity
- Completion Patterns
- Concept Difficulty
- Learning Retention
- Career Readiness

11. Experimental Framework

Every new idea should be validated experimentally.

Research Process

Observation

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Hypothesis

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Prototype

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User Experiment

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Data Collection

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Analysis

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Iteration

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Deployment

No algorithm should become part of Bloom without measurable improvement.

12. Research Metrics

Bloom measures research success using learning-centered metrics.

Examples

Learning Efficiency

- Time to concept mastery
- Time spent searching vs learning

Learning Quality

- Knowledge retention
- Concept understanding
- Skill progression

Recommendation Quality

- Acceptance rate
- Completion rate
- Satisfaction
- Recommendation confidence

Career Outcomes

- Portfolio completion

- Interview readiness
 - Job placement (where measurable)
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13. Data Collection Principles

Research should respect user privacy.

Guidelines

- Collect only necessary information.
 - Prefer anonymized analytics.
 - Be transparent about data usage.
 - Allow users to opt out.
 - Never sell personal learning data.
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14. Collaboration Strategy

Bloom should collaborate with:

Academic Researchers

- Learning Sciences
- AI
- Human-Computer Interaction
- Educational Technology

Industry

- EdTech companies
- AI companies
- Universities
- Open-source communities

Learners

- Continuous feedback
 - Beta programs
 - Community testing
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15. Future Research Questions

Examples of long-term investigations:

- Can AI estimate concept mastery without formal tests?
 - Can learning styles be inferred from interaction patterns?
 - Can career success be predicted from learning trajectories?
 - Can Bloom recommend the optimal time to switch topics?
 - Can AI automatically construct skill graphs from public knowledge?
 - How can Bloom reduce learner burnout?
 - How should collaborative learning be modeled?
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16. Publications & Open Research

Where appropriate, Bloom should contribute back to the research community through:

- Technical blogs
- Whitepapers
- Open datasets (privacy-preserving)
- Open-source tools
- Conference talks
- Research collaborations

Sharing selected findings can improve credibility and attract collaborators while protecting genuinely proprietary intellectual property.

17. Research Principles

Every research initiative should satisfy these principles:

1. Improve learner outcomes.
 2. Be measurable and reproducible.
 3. Respect learner privacy.
 4. Be grounded in evidence.
 5. Balance innovation with practicality.
 6. Produce insights that can improve the product.
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18. Long-Term Research Vision

Bloom's long-term ambition is not simply to build another recommendation system.

It is to establish a new field of **Learning Intelligence**.

Learning Intelligence combines:

- Artificial Intelligence
- Learning Science
- Knowledge Graphs
- Educational Psychology
- Human-Computer Interaction
- Recommendation Systems
- Data Science

The objective is to understand how humans acquire knowledge and use that understanding to help every learner navigate the world's educational resources more effectively.

Research Summary

Research is a strategic pillar of Bloom. Every product feature should be informed by evidence, experimentation, and continuous learning.

Bloom's research mission is to answer one enduring question:

How can we help every learner discover, understand, retain, and apply the right knowledge at the right moment in their journey?

By continuously investigating this question, Bloom evolves from a recommendation product into a research-driven Learning Intelligence Platform capable of improving lifelong learning at global scale.